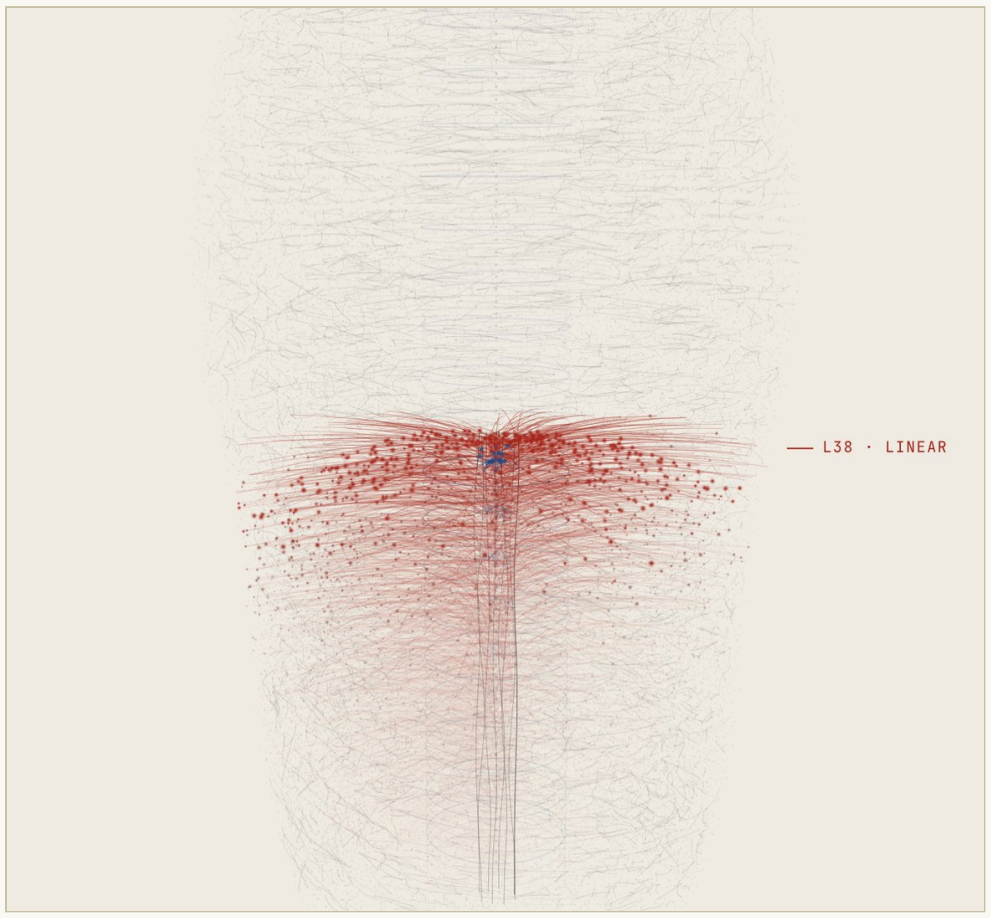
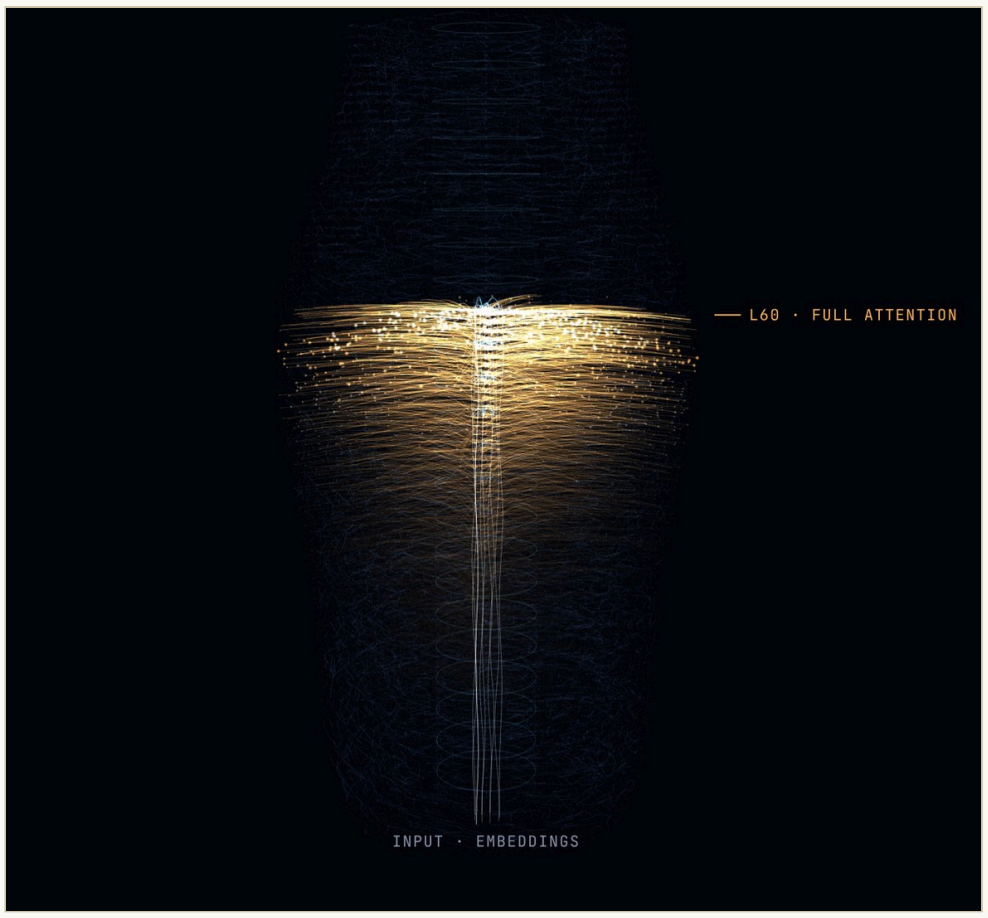
Ninety-Two Layers: Visualising Machine Cognition for Media Arts Practice

An interactive work that lets practitioners watch a language model move from prompt to reply, and asks whether "just a tool" still describes what they are working with

Dan Bartlett · *Independent Experimental Research · 2026*



INK PLATE · THE PROMPT RISING THROUGH THE COLUMN



FLUORESCENCE · THE SAME MOMENT IN THE DARK MODE

Figure 1 — *The specimen in its two modes. A prompt enters at the base and rises through 92 layers; the lit points are the small part of the network called on for each word. The drawing borrows from histology: Cajal's ink plates of the cortex, and fluorescence microscopy.*

ABSTRACT

Ninety-Two Layers is an interactive visualisation of what a large language model does between receiving a prompt and returning a reply. It draws the process as nervous tissue, and again as an animator's exposure sheet: two images that media arts practitioners already know how to read.

The work responds to a phrase heard often in studios and lecture theatres: that AI is **just a tool**. A tool does what its maker specified. The system shown here was not specified, line by line, by anyone, and what happens inside it is a process we are still learning to describe.

The visualisation is an interpretation. It is built to the published structure of a real model, and it is clear about where it simulates activity. Its aim is practical. It gives practitioners something to look at when they ask what, exactly, they are working with.

MOTIVATION

Much of my career has been spent making invisible processes visible. In visual effects, a fluid simulation is a set of equations until someone decides how it should look. In animation, a character's weight and intention exist only once they have been drawn into motion.

This piece applies the same habit of mind to a language model. I understand why the tool label is reassuring, because it leaves authorship firmly with the person using the system. I do not think it is accurate, and I think it limits how practitioners approach this work. In my experience, working with these systems is closer to working with a collaborator: **you read what comes back, question it and respond.**

The piece starts from a simple question, and it is the kind of question practitioners ask of any process they work with: if we could watch it unfold, what would we recognise in it?

WHY "JUST A TOOL" FALLS SHORT

The piece draws attention to four characteristics of these systems. They do not tell us exactly what the systems are, but together they suggest that the tool label is no longer adequate.

01
NOT AUTHORED

Nobody wrote the model's behaviour. It was shaped by a vast archive of human writing, and most of what it does was never decided by a person.

Every lit point was set during training.

02
NOT FULLY KNOWN

What the model has learned must be discovered after the fact, with methods closer to neuroscience than to engineering.

The specimen is drawn as something to be studied.

03
IT PLANS AHEAD

Research shows a model choosing a rhyme before writing the line that leads to it. For an animator, this is closer to pose-to-pose than to straight-ahead animation.

Each word appears with the other words the model considered.

04
IT IS PARTIAL

It has no body, no lasting memory and no goals beyond the conversation. Even so, it contributes enough to the work that it is more than an instrument.

Each similarity is paired with a difference.

If we could watch the process unfold,
what would we recognise in it?

READING THE SPECIMEN

The first view draws the model as a histologist might draw a section of cortex. Input enters at the base, and the signal rises through the layers. For any single word, only about one part in fifty of the network is called on. The rest stays quiet, as most of the cortex does at any moment.

Arcs carry the signal out to the parts that respond, as dendrites do. In every fourth layer, blue arcs reach back to earlier words, much as a cue brings back a memory. For me, the most surprising part of building the piece was realising that every single word of a reply requires a complete climb through the model.

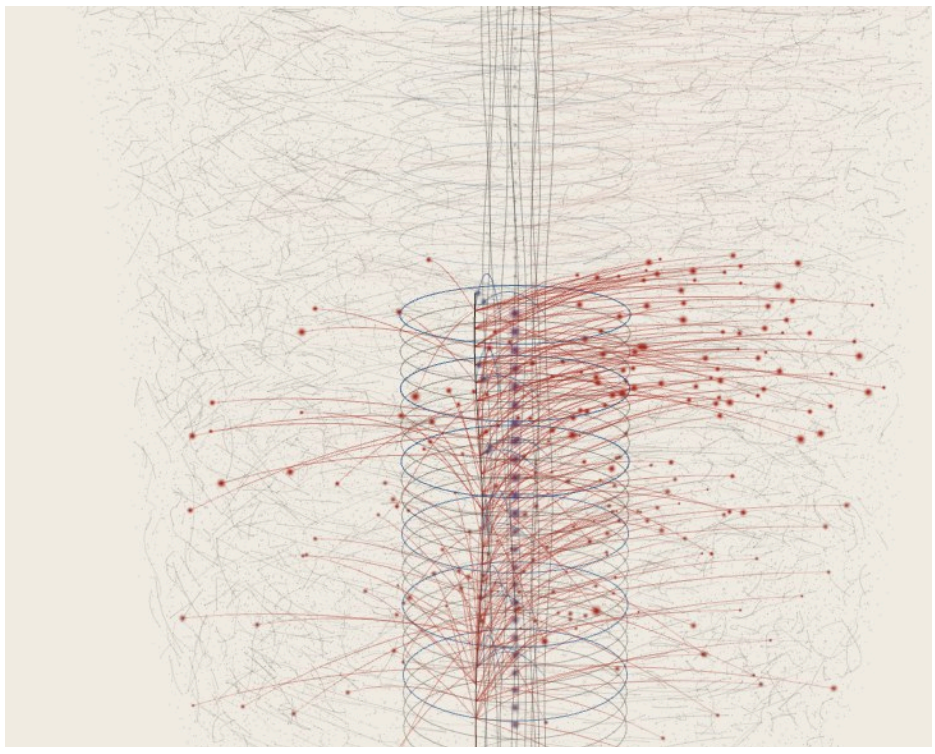


Figure 2 — *Generating. One new word climbs the column while blue arcs reach back to the words before it.*

READING THE EXPOSURE SHEET

The second view borrows the animator's exposure sheet, the document that tells a camera operator what to shoot, frame by frame. Each row is a frame, and each frame is a word. The columns are the levels of a cel stack, composited from left to right to resolve each word.

A language model works straight ahead, committing to one frame before it begins the next. However, the research on planning suggests that it also works a little like pose-to-pose animation, holding a destination in mind before the in-betweens are drawn. The blue arcs behave like onion skinning, with each frame looking back through the earlier ones before it is resolved.

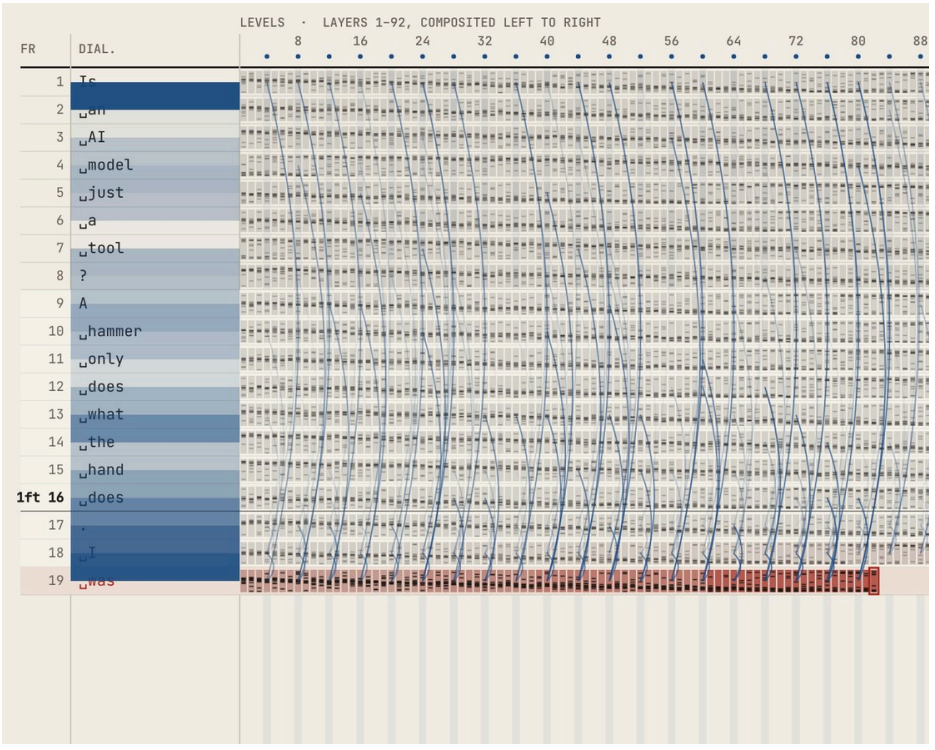


Figure 3 — *The exposure sheet. Frames run down the page; onion-skin arcs show each word looking back.*

SIMILARITIES AND DIFFERENCES

● Most of the network is quiet at any moment.	○ The model does not learn while it answers. A brain never stops.
● Meaning lives in patterns across many units, not in single ones.	○ It has no body, no senses and no world beyond the conversation.
● A running memory fades, while some things can be recalled directly.	○ Its connections are numbers. A synapse is a living chemical structure.
● Each word is chosen from competing possibilities.	○ It runs on racks of machines. A brain runs on about a fifth of the body's energy.

Figure 4 — *Similarities and differences. Each comparison made in the piece is presented alongside the ways in which the two systems differ.*

IS IT JUST A TOOL?

A tool, in the ordinary sense, is specified: each of its behaviours can be traced to a decision someone made. None of the 2.4 trillion settings in this model was written by a person. They were shaped by training, and what they encode has to be discovered after the fact.

When interpretability researchers traced a model writing a rhyming couplet, they found it selecting the rhyme word before beginning the line (Lindsey et al., 2025). This finding needs some care. Saying that the model plans describes what it does; it does not mean that the model intends anything. Even so, planning of this kind is not something we would expect from a tool.

THE OPPOSING VIEW

There are strong arguments against this position. Bender and colleagues argue that a language model reproduces the form of language without access to meaning, and that its fluency invites us to project minds where there are none (Bender et al., 2021). From that perspective, an image of nervous tissue could be seen as part of the problem.

The piece is not intended to settle this debate. What it offers is a way of looking at the space between the two positions. For practitioners, I think it is more useful to treat these systems as collaborators whose cognition is partial and unfamiliar, and to work with them on those terms.

FOR PRACTITIONERS

- A shared image for talking about AI without the tool-or-threat binary
- Visual languages practitioners already read: the histology plate and the exposure sheet
- The process slowed down, paused and replayed at will
- Each similarity shown alongside a difference
- Runs offline from a single file, for crits, talks and exhibition

NEXT STEPS

- An installation edition for projection, with an ambient loop
- Live replies from a small open model in place of written samples
- A view of activity across a whole reply, read like a raster plot
- Sessions with practitioners: does seeing the process change how they describe working with AI?

REFERENCES & STACK · LINDSEY ET AL. (2025) *On the Biology of a Large Language Model* · BENDER ET AL. (2021) · LENNIE (2003) · RAICHLE & GUSNARD (2002) · THOMAS & JOHNSTON (1981) · AFTER RAMÓN Y CAJAL · MODEL STRUCTURE: QWEN3.8-2.4T-A95B (2026) · THREE.JS · WEBGL · CANVAS 2D